

Working Capital Decisions



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Working Capital

- Measure of a company's operational efficiency and short-term financial health.
- Calculated as the difference between current assets and current liabilities.
- Ensures a company has sufficient resources to continue its operations and meet its short-term obligations.

Components of Working Capital

➤ Current Assets

- Cash and cash equivalents,
- accounts receivable,
- inventory, and
- marketable securities

➤ Current Liabilities

- Accounts payable,
- short-term debt,
- accrued liabilities, and
- interest payable on outstanding debts

Importance of Working Capital

Adequate working capital ensures that a company can:

- meet its short-term obligations,
- maintain smooth operations, and
- avoid financial difficulties.

Insufficient working capital may lead to operational disruptions.

Determinants

Working Capital Management

Managing the level of individual current assets and current liabilities



Working Capital Financing

Methods and sources a company can use to fund its day-to-day operations



Working Capital Financing



Permanent working capital

- Financial foundation of a company's everyday activities
- Always needed irrespective of sales fluctuation
- Should be financed by the long-term sources

Temporary working capital

- Covers immediate operational needs
- Additional funds required to meet short-term or seasonal variations in production and sales
- May be financed by the short-term sources of finance

Sources of Working Capital Financing

Short-term Sources

TRADE CREDIT

CASH CREDIT

OVERDRAFT

DISCOUNTING

Long-term Sources

EQUITY

LONG-TERM DEBT

Short-term Sources

Trade Credit

- Extended by suppliers, allowing the business to purchase goods or services and pay for them later.
- Depends on various factors like the buyer's reputation, financial position, business volume, and degree of competition
- Advantages:
 - Cost-effective source of financing
 - No immediate cash outflow
 - Helps businesses manage their cash flow more effectively
- Risks:
 - Overreliance on trade credit can strain supplier relationships if payments are delayed.

Short-term Sources

Trade Credit

$$\frac{2}{10} \text{ Net } 60$$

Credit period = 60 days

If paid within 10 days = 2% discount

Cost of not availing discount:

$$Cost = \left[\frac{\text{Discount}}{(100 - \text{Discount})} \times 100 \times \frac{365 \text{ days}}{(\text{Credit Period} - \text{Discount Period})} \right]$$

$$Cost = \left[\frac{2}{(100 - 2)} \times 100 \times \frac{365 \text{ days}}{(60 \text{ days} - 10 \text{ days})} \right]$$

$$Cost = 14.9\%$$

Short-term Sources

Cash Credit (CC)

- Short-term loan offered to businesses by financial institutions like banks.
- Repayment period of up to 12 months.
- Borrowing limited determined based on credit history and financial stability.
- Requires to hypothecate stocks and receivables.
- Interest rate is charged only on the amount withdrawn and not on the total sanctioned limit.
- Interest rate charged on cash credit is high compared to traditional loans.

Short-term Sources

Overdraft (OD)

- Agreement with a bank to withdraw more money than it has in its account up to a specified limit.
- Bridge cash flow gaps during lean periods.
- Limit is sanctioned based on financials and security deposits
- Interest rates are only applicable to the amount utilised from the total sanctioned limit.
- Rate of interest of an Overdraft is higher than that of a Cash Credit.

Short-term Sources

Discounting

- Known as 'bill discounting' or 'invoice discounting'
- Advance selling of a bill to a financial institution before it is due to be paid.
- Financial institute charges some amount as commission called a 'discount'
- Discount is treated as the cost of raising funds through this method
- Provides immediate cash flow by converting outstanding invoices into cash.

Short-term Sources

Other Sources

- Commercial Paper
 - US - maximum 270 days
 - India - maximum 1 year
- Line of Credit
- Bank Guarantee
- Advances from Customers
- Debt Securitisation
 - Mortgage-backed securities (MBS)
 - Collateralized debt obligations (CDO)

Security for Short-term Financing

➤ Pledge of Accounts receivables

Using outstanding invoices to secure a loan or credit line

➤ Charge on Inventory

Using inventory as security to secure a loan.

Pledge of Accounts receivables

- Using receivables as security to secure a loan or line of credit from a lender
- The lender assesses the quality and age of the receivables.
- **Advance rate** - Percentage of the collateral's value, reflecting the maximum loan amount that a lender is willing to extend to a borrower.
- Depending on the agreement, the borrower may continue to collect receivables or the lender might take control of the collection process.
- If the company defaults, the lender has the right to collect the receivables directly from the customers.

Security for Short-term Financing

Charge on Inventory

- Using a company's inventory as security to secure a loan.
- The lender assesses the value of the inventory to determine how much financing they are willing to provide.
- If the borrower defaults, the lender has a claim on the inventory and can sell it to recover the loan amount.
- Floating Charge
 - Company is allowed to use the inventory in its ordinary course of business until an event like default occurs.
- Fixed Charge
 - Specific inventory items identified at the time of the loan agreement which cannot be sold or disposed without the lender's permission.

Security for Short-term Financing

Working capital Estimation

Factors Influencing Working Capital Requirements

Nature of Business

Business Cycle

Market Conditions

Inventory Policy

Production Process & Length

Availability of Credit from Suppliers

Price-level Changes

Credit Policy

Working capital Estimation

Operating Cycle Method

- Operating cycle is the time taken to convert raw materials into cash through sales.
- This method helps in finding the funds held-up in different stages of operations and the time required to recover.

Regression Analysis Method

- Statistical method used to estimate the relationship between variables.
- Working Capital is estimated by establishing relationship between working capital, sales and other factors.

Operating Cycle Method

Operating cycle is the time taken to convert raw materials into cash through sales.

Working Capital Requirement (WCR):

$$WCR = \frac{\text{Operating Cycle in days}}{365} \times \text{Total Annual Operating Expenses}$$

Operating Cycle Components:

- **Inventory Conversion Period (ICP):** Time taken to convert raw materials into finished goods and then sell them.
- **Receivables Conversion Period (RCP):** Time taken to collect cash from customers after a sale.
- **Payables Deferral Period (PDP):** Time the company takes to pay its suppliers.

$$\text{Net Operating Cycle} = \text{ICP} + \text{RCP} - \text{PDP}$$

Operating Cycle Method

Operating Cycle Components:

Inventory Conversion Period (ICP):

$$ICP = \frac{\text{Average Inventory}}{\left(\frac{\text{Cost of Goods Sold}}{365}\right)}$$

Receivables Conversion Period (RCP):

$$RCP = \frac{\text{Average Receivables}}{\left(\frac{\text{Net Credit Sales}}{365}\right)}$$

Payables Deferral Period (PDP):

$$PDP = \frac{\text{Average Payables}}{\left(\frac{\text{Cost of Goods Sold}}{365}\right)}$$

Question 1:

A company has provided the following financial data for the past year:

The average inventory is \$200,000, the Cost of Goods Sold (COGS) is \$1,000,000, the average receivables amount to \$150,000, and the net credit sales are \$900,000.

Additionally, the company has average payables of \$100,000, and the total annual operating expenses amount to \$800,000. Using the Operating Cycle Method, calculate the company's Working Capital Requirement.

Average Inventory = \$200,000

Cost of Goods Sold (COGS) = \$1,000,000

Average Receivables = \$150,000

Net Credit Sales = \$900,000

Average Payables = \$100,000

Total Annual Operating Expenses = \$800,000

Operating Cycle Components:

Inventory Conversion Period (ICP):

$$ICP = \frac{\text{Average Inventory}}{\left(\frac{\text{Cost of Goods Sold}}{365}\right)} = \frac{200,000}{\left(\frac{1,000,000}{365}\right)} = 73$$

Receivables Conversion Period (RCP):

$$RCP = \frac{\text{Average Receivables}}{\left(\frac{\text{Net Credit Sales}}{365}\right)} = \frac{150,000}{\left(\frac{900,000}{365}\right)} = 60.83$$

Payables Deferral Period (PDP):

$$PDP = \frac{\text{Average Payables}}{\left(\frac{\text{Cost of Goods Sold}}{365}\right)} = \frac{100,000}{\left(\frac{1,000,000}{365}\right)} = 36.50$$

$$\text{Operating Cycle} = ICP + RCP - PDP$$

$$\text{Operating Cycle} = 73 + 60.83 - 36.5$$

$$\text{Operating Cycle} = 97.33$$

Working Capital Requirement (WCR):

$$WCR = \frac{\text{Operating Cycle in days}}{365} \times \text{Total Annual Operating Expenses}$$

$$WCR = \frac{97.33}{365} \times 800,000$$

$$WCR = 213,333$$

Operating Cycle Method

Operating cycle is the time taken to convert raw materials into cash through sales.

Working Capital Requirement (WCR):

$$WCR = \left(\frac{\text{Cash Cycle} \times \text{Annual Operating Expenses}}{365} \right) + \text{Desired Cash}$$

$$\text{Cash Cycle} = R + W + F + D - C$$

R = Average period the funds will be held-up in raw material form.

$$R = \frac{\text{Average Stock of raw material}}{\text{Raw material consumption per day}}$$

W = Average period the funds will be held-up in manufacturing process.

$$W = \frac{\text{Average Stock of Work in progress}}{\text{Cost of production per day}}$$

Operating Cycle Method

Operating cycle is the time taken to convert raw materials into cash through sales.

F = Average period the funds will be held-up in the form of finished goods.

$$F = \frac{\text{Average Stock of Finished Goods}}{\text{Average Cost of Goods Sold per day}}$$

D = Average credit period granted to customers.

$$D = \frac{\text{Average Accounts Receivables}}{\text{Average Credit Sales per day}}$$

C = Average credit period granted by suppliers.

$$C = \frac{\text{Average Accounts Payables}}{\text{Average Credit Purchases per day}}$$

Question 3:

The following information is available for a company that expects an annual operating expenses of \$1,200,000. Calculate the operating cycle duration and working capital requirement of the company.

Particulars	Amount (in lakhs)
Average stock of raw materials and stores	150
Average work-in-process inventory	250
Average finished goods inventory	170
Average accounts receivable	250
Average accounts payable	160
Average raw materials and stores consumed per day	10
Average cost of production per day	20
Average cost of goods sold per day	17
Average Sales per day	20
Average credit purchases per day	8

Cash Conversion Cycle:

$$R = \frac{\text{Average Stock of raw material}}{\text{Raw material consumption per day}} = \frac{150}{10} = 15$$

$$W = \frac{\text{Average Stock of Work in progress}}{\text{Cost of production per day}} = \frac{250}{20} = 12.5$$

$$F = \frac{\text{Average Stock of Finished Goods}}{\text{Average Cost of Goods Sold per day}} = \frac{170}{17} = 10$$

$$D = \frac{\text{Average Accounts Receivables}}{\text{Average Credit Sales per day}} = \frac{250}{20} = 12.5$$

$$C = \frac{\text{Average Accounts Payables}}{\text{Average Credit Purchases per day}} = \frac{160}{8} = 20$$

$$\text{Cash Cycle} = R + W + F + D - C$$

$$\text{Cash Cycle} = 15 + 12.5 + 10 + 12.5 - 20$$

$$\text{Cash Cycle} = 30$$

Working Capital Requirement (WCR):

$$WCR = \left(\frac{\text{Cash Cycle} \times \text{Annual Operating Expenses}}{365} \right)$$

$$WCR = \frac{30 \times 1,200,000}{365}$$

$$WCR = 98,630$$

Question 4:

The following information is forecasted by NovaTech Solutions Ltd. for the year ending 31st March 2025:

Based on the data provided, calculate the Cash Operating Cycle.

Particulars	Balance as at 1st April 2024	Balance as at 31st March 2025
Raw material	45,000	65,000
Work-in-progress	35,000	51,300
Finished goods	60,181	70,200
Debtors	1,12,123	1,35,000
Creditors	50,079	70,500
Annual purchase of raw material (all credit)		4,00,000
Annual cost of production		7,50,000
Annual cost of goods sold		9,15,000
Annual operating cost		9,50,000
Annual sales (all credit)		11,00,000

Particulars	Calculations	Amount (in lakhs)
Average stock of raw materials	$(45,000 + 65,000) / 2$	55,000.00
Average work-in-process inventory	$(35,000 + 51,300) / 2$	43,150.00
Average finished goods inventory	$(60,181 + 70,200) / 2$	65,190.50
Average accounts receivable	$(112,123 + 135,000) / 2$	1,23,561.50
Average accounts payable	$(50,079 + 70,500) / 2$	60,289.50
Average raw material consumption per day	$(400,000 + 45,000 - 65,000) / 365$	1,040.10
Average cost of production per day	$750,000 / 365$	2,054.79
Average cost of goods sold per day	$915,000 / 365$	2,506.85
Average Credit Sales per day	$1,100,000 / 365$	3,013.70
Average credit purchases per day	$400,000 / 365$	1,095.89

Cash Conversion Cycle:

$$R = 55,000 / 1041.10 = 52.83$$

$$W = 43,150 / 2054.79 = 21.00$$

$$F = 65,190.50 / 2,506.85 = 26.00$$

$$D = 123,561.50 / 3,013.70 = 41.00$$

$$C = 60,289.50 / 1,095.89 = 55.01$$

$$\text{Cash Cycle} = R + W + F + D - C$$

$$\text{Cash Cycle} = 52.83 + 21.00 + 26.00 + 41.00 - 55.01$$

$$\text{Cash Cycle} = 85.82$$

Number of Operating Cycles in a year:

$$\text{Number of Operating Cycles} = 365 / 85.82 = 4.25$$

Working Capital Requirement (WCR):

$$WCR = \left(\frac{\text{Cash Cycle} \times \text{Annual Operating Expenses}}{365} \right).$$

$$WCR = (85.82 \times 950,000) / 365$$

$$WCR = 223,367$$

Question 5:

Calculate the working capital requirement if the sales of the company are Rs. 2,000 lakhs, and the cash required for contingency is Rs. 4.75 lakhs.

Additional information:

1. Purchase of raw materials and stores is Rs. 810 lakhs
2. Conversion cost is Rs. 600 lakhs
3. Selling, distribution and financial cost is Rs. 240 lakhs

Particulars	Opening Balance (in lakhs)	Closing Balance (in lakhs)
Raw material	230	250
Work-in-progress	50	52
Finished goods	260	300
Debtors	380	450
Creditors	250	300

Annual raw material consumption = Purchase of raw material + Opening Balance - Closing Balance

Annual raw material consumption = $810 + 230 - 250 = 790$

Annual cost of production = Annual raw material consumption + Conversion Cost + Opening Balance - Closing Balance

Annual cost of production = $790 + 600 + 50 - 52 = 1388$

Annual Cost of Goods Sold = Annual cost of production + Opening Balance - Closing Balance

Annual Cost of Goods Sold = $1388 + 260 - 300 = 1348$

Annual Credit Sales = **2000**

Annual credit purchases = Purchase of raw material and stores

Annual credit purchases = **810**

Particulars	Calculations	Amount (in lakhs)
Average stock of raw materials	$(230 + 250) / 2$	240.00
Average work-in-process inventory	$(50 + 52) / 2$	51.00
Average finished goods inventory	$(260 + 300) / 2$	280.00
Average accounts receivable	$(380 + 450) / 2$	415.00
Average accounts payable	$(250 + 300) / 2$	275.00
Average raw material consumption per day	$790 / 365$	2.16
Average cost of production per day	$1388 / 365$	3.80
Average cost of goods sold per day	$1348 / 365$	3.69
Average Credit Sales per day	$2000 / 365$	5.48
Average credit purchases per day	$810 / 365$	2.22

Cash Conversion Cycle:

$$R = 240 / 2.16 = 111.11$$

$$W = 51 / 3.80 = 13.42$$

$$F = 280 / 3.69 = 75.88$$

$$D = 415 / 5.48 = 75.73$$

$$C = 275 / 2.22 = 123.87$$

$$\text{Cash Cycle} = R + W + F + D - C$$

$$\text{Cash Cycle} = 53.05 + 21.00 + 26.00 + 41.00 - 55.00$$

$$\text{Cash Cycle} = 152.27$$

Annual Operating Cost:

$$\text{Annual Operating Cost} = 1348 + 240 = 1588$$

Working Capital Requirement (WCR):

$$WCR = \left(\frac{\text{Cash Cycle} \times \text{Annual Operating Expenses}}{365} \right) + \text{Desired Cash}$$

$$WCR = (152.27 \times 1588 / 365) + 4.75$$

$$WCR = 667.23$$

Regression Analysis Method

Statistical method used to estimate the relationship between variables

01

Data Collection

Collect historical data on sales and working capital.

02

Prepare Regression Equation

Run a regression analysis to identify the correlation between sales and working capital.

Regression Equation: $WC = a + b (\text{Sales})$

03

Estimate Net Working Capital

Use the regression equation to forecast future working capital needs.

Correlation Coefficient (r):

Measures the strength and direction of the linear relationship between the two variables.

Slope (b):

Calculates the amount of change in dependence variable for each unit change in independent variable.

Intercept (a):

Signifies the value of dependent variable when the value of slope is 0.

Method 1:

$$r = \frac{\sum (x_i - \bar{x}) (y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

$$\text{Slope (b)} = r \times \frac{\text{Standard deviation of } y}{\text{Standard deviation of } x}$$

$$\text{Intercept (a)} = \bar{y} - b \times \bar{x}$$

Method 2:

$$\text{Slope (b)} = \frac{N \sum xy - \sum x \sum y}{N \sum x^2 - (\sum x)^2}$$

$$\text{Intercept (a)} = \bar{y} - b \times \bar{x}$$

Question 6:

A retail company has collected the following historical data over the last five years:

Based on the following information:

- 1. Derive the regression equation in the form of $WC = a + b(\text{Sales})$.
- 2. Using the regression equation, estimate the working capital required if the projected sales for the upcoming year are \$22 million.

Year	Sales (\$ millions)	Working Capital (\$ millions)
1	10	2.5
2	12	3.0
3	14	3.5
4	18	4.0
5	21	4.5

Year	Sales (\$ millions) (x)	Working Capital (\$ millions) (y)	x^2	xy
1	10	2.5	100	25
2	12	3	144	36
3	14	3.5	196	49
4	18	4	324	72
5	21	4.5	441	94.5
Total	75	17.5	1205	276.5

Mean Sales (x) = Sum of Sales / Number of data points = 75 / 5 = 15

Mean WC (y) = Sum of WC / Number of data points = 17.5 / 5 = 3.5

$$Slope (b) = \frac{N \sum xy - \sum x \sum y}{N \sum x^2 - (\sum x)^2} = \frac{5 \times 276.5 - 75 \times 17.5}{5 \times 1205 - (75)^2} = 0.175$$

$$Intercept (a) = \bar{y} - b \times \bar{x} = 3.5 - 0.175 \times 15 = 0.875$$

Regression Equation:

$$WC = a + b \text{ (Sales)}$$

$$WC = 0.875 + 0.175 \text{ (Sales)}$$

Estimated working capital required for projected sales of \$22 million is:

$$WC = 0.875 + 0.175 \times 22$$

$$WC = 4.725 \text{ million}$$

Question 7:

Silverline Manufacturing Ltd. is analyzing its working capital requirements using regression analysis. The company has gathered the following data for the past five years:

Using the data provided, perform a regression analysis to determine the relationship between sales and working capital. Estimate the working capital requirement if Silverline Manufacturing Ltd. expects its sales to reach ₹800 lakhs in 2024.

Year	Sales (₹ in lakhs)	Working Capital (₹ in lakhs)
2019	500	120
2020	550	135
2021	600	145
2022	650	160
2023	700	175

Year	Sales (₹ in lakhs) (x)	Working Capital (₹ in lakhs) (y)	x - Mean of x	y - Mean of y	(x - Mean x)^2	(y - Mean y)^2	(x - Mean x) x (y - Mean y)
2019	500	120	-100	-27	10000	729	2700
2020	550	135	-50	-12	2500	144	600
2021	600	145	0	-2	0	4	0
2022	650	160	50	13	2500	169	650
2023	700	175	100	28	10000	784	2800
Total	3000	735	0	0	25000	1830	6750

Mean of Sales (x) = Sum of Sales / Number of data points = 3000 / 5 = 600

Mean of WC (y) = Sum of WC / Number of data points = 735 / 5 = 147

$$SD(x) = \sqrt{\frac{\sum(x - \bar{x})^2}{(n - 1)}} = \sqrt{\frac{25000}{(5 - 1)}} = 79.06$$

$$SD(y) = \sqrt{\frac{\sum(y - \bar{y})^2}{(n - 1)}} = \sqrt{\frac{1830}{(5 - 1)}} = 21.39$$

Correlation Coefficient (r):

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}} = \frac{6750}{\sqrt{25000 \times 1830}} = 0.998$$

Slope (b):

$$\text{Slope (b)} = r \times \frac{SD(y)}{SD(x)} = 0.998 \times \frac{21.39}{79.06} = 0.27$$

Intercept (a):

$$\text{Intercept (a)} = \bar{y} - b \times \bar{x} = 147 - 0.27 \times 600 = -15$$

Regression Equation:

$$WC = a + b (\text{Sales})$$

$$WC = -15 + 0.27 (\text{Sales})$$

Estimated working capital required for projected sales of ₹800 lakhs is:

$$WC = -15 + 0.27 \times 800$$

$$WC = 201 \text{ million}$$

Working Capital Leverage

- Measure of how sensitive a company's profitability (EBIT) is to changes in working capital.
- $WCL = \% \text{ change in EBIT} / \% \text{ change in working capital}$
- Higher leverage indicates that small changes in working capital have a significant impact on profitability
- Effective management of working capital leverage ensures that the company can optimize returns while minimizing financial risks.

QUESTIONS?

